CSE 201 Practice Lab 03

Algorithm Analysis - 30 Question Practice Bank

Questions	Format	Score
30	Java answer worksheet	100 points

Instructions

For every question, determine the asymptotic running-time bound and write only the required expression inside the blank `answer` string. Do not change a method signature or its print statement.

Allowed answer forms: `1`, `logn`, `n`, `nlogn`, `n^2`, `n^3`, `2^n`, `n+m`, and `n*m`.

Workflow: complete the Java file first, then run `PracticeLab03Tester` for the detailed 100-point report.

Question Source Code

```
import java.util.ArrayList;

public class PracticeLab03 {

    // CSE 201 - Data Structures | Practice Lab 03: Algorithm Analysis
    //
    // For every question, write your Big-O answer inside the quotes of answer.
    // Allowed forms: 1, logn, n, nlogn, n^2, n^3, 2^n, n+m, n*m.
    // Do not change method signatures or print statements.
```

Question 01

```
    public static long question01(int[] a) {
        String answer = "";
        System.out.printf("Q01: O(%s)%n", answer);

        if (a.length == 0) {
            return 0;
        }

        return a[0] + a[a.length - 1];
    }
```

Question 02

```
    public static long question02(int[] a) {
        String answer = "";
        System.out.printf("Q02: O(%s)%n", answer);
```

```
    long sum = 0;

    for (int value : a) {
        sum += value;
    }

    return sum;
}
```

Question 03

```
public static long question03(int[] a) {
    String answer = "";
    System.out.printf("Q03: O(%s)%n", answer);

    long sum = 0;

    for (int i = 0; i < a.length; i++) {
        for (int j = 0; j <= i; j++) {
            sum += a[j];
        }
    }

    return sum;
}
```

Question 04

```
public static long question04(int[] a) {
    String answer = "";
    System.out.printf("Q04: O(%s)%n", answer);

    long sum = 0;

    for (int i = 0; i < a.length; i++) {
        for (int j = 0; j < a.length; j++) {
            sum += a[i] * a[j];
        }
    }

    return sum;
}
```

Question 05

```
public static long question05(int[] a) {
    String answer = "";
    System.out.printf("Q05: O(%s)%n", answer);

    long sum = 0;

    for (int i = 0; i < a.length; i++) {
        for (int j = 0; j < a.length; j++) {
            for (int k = 0; k < a.length; k++) {
                int index = (i + j + k) % a.length;
                sum += a[index];
```

```
            }
        }
    }

    return sum;
}
```

Question 06

```
public static long question06(int[] a) {
    String answer = "";
    System.out.printf("Q06: O(%s)%n", answer);

    if (a.length == 0) {
        return 0;
    }

    long sum = 0;
    int size = a.length;

    while (size > 1) {
        int index = size % a.length;
        sum += a[index];
        size /= 2;
    }

    return sum;
}
```

Question 07

```
public static long question07(int[] a) {
    String answer = "";
    System.out.printf("Q07: O(%s)%n", answer);

    long sum = 0;

    for (int i = 0; i < a.length; i++) {
        int size = a.length;

        while (size > 1) {
            sum += a[i];
            size /= 2;
        }
    }

    return sum;
}
```

Question 08

```
public static long question08(int[] a) {
    String answer = "";
    System.out.printf("Q08: O(%s)%n", answer);

    return branch(a, 0);
```

```
}

private static long branch(int[] a, int index) {
    if (index == a.length) {
        return 1;
    }

    long left = branch(a, index + 1);
    long right = branch(a, index + 1);

    return left + right;
}
```

Question 09

```
public static long question09(int[] a, int[] b) {
    String answer = "";
    System.out.printf("Q09: O(%s)%n", answer);

    long sum = 0;

    for (int x : a) {
        for (int y : b) {
            sum += x + y;
        }
    }

    return sum;
}
```

Question 10

```
public static long question10(int[] a, int[] b) {
    String answer = "";
    System.out.printf("Q10: O(%s)%n", answer);

    long sum = 0;

    for (int x : a) {
        sum += x;
    }

    for (int y : b) {
        sum += y;
    }

    return sum;
}
```

Question 11

```
public static long question11(int[] a) {
    String answer = "";
    System.out.printf("Q11: O(%s)%n", answer);

    long pairs = 0;
```

```
        for (int i = 0; i < a.length; i++) {
            for (int j = i + 1; j < a.length; j++) {
                pairs += a[i] + a[j];
            }
        }

        return pairs;
    }
```

Question 12

```
    public static long question12(int[] a) {
        String answer = "";
        System.out.printf("Q12: O(%s)%n", answer);

        long sum = 0;

        for (int i = 0; i < a.length; i += 2) {
            sum += a[i];
        }

        return sum;
    }
```

Question 13

```
    public static long question13(int[] a) {
        String answer = "";
        System.out.printf("Q13: O(%s)%n", answer);

        long sum = 0;
        int width = 1;

        while (width < a.length) {
            sum += width;
            width *= 2;
        }

        return sum;
    }
```

Question 14

```
    public static long question14(int[] a) {
        String answer = "";
        System.out.printf("Q14: O(%s)%n", answer);

        long sum = 0;

        for (int i = 0; i < a.length; i++) {
            for (int j = 0; j < a.length; j++) {
                sum += a[j];
            }
        }

        return sum;
```

```
}
```

Question 15

```
public static long question15(int[] a) {
    String answer = "";
    System.out.printf("Q15: O(%s)%n", answer);

    return linearRecursion(a, 0);
}

private static long linearRecursion(int[] a, int index) {
    if (index == a.length) {
        return 0;
    }

    return a[index] + linearRecursion(a, index + 1);
}
```

Question 16

```
public static long question16(int[] a) {
    String answer = "";
    System.out.printf("Q16: O(%s)%n", answer);

    long sum = 0;
    int left = 0;
    int right = a.length - 1;

    while (left <= right) {
        sum += a[left];
        left++;
        right--;
    }

    return sum;
}
```

Question 17

```
public static long question17(int[] a) {
    String answer = "";
    System.out.printf("Q17: O(%s)%n", answer);

    long sum = 0;
    int i = 1;

    while (i < a.length) {
        for (int j = 0; j < i; j++) {
            sum += j;
        }

        i *= 2;
    }

    return sum;
```

```
}
```

Question 18

```
public static long question18(int[] a) {
    String answer = "";
    System.out.printf("Q18: O(%s)%n", answer);

    long sum = 0;
    int step = 1;

    while (step <= a.length) {
        for (int value : a) {
            sum += value;
        }

        step *= 2;
    }

    return sum;
}
```

Question 19

```
public static long question19(int[] a) {
    String answer = "";
    System.out.printf("Q19: O(%s)%n", answer);

    String text = "";

    for (int value : a) {
        text += value;
    }

    return text.length();
}
```

Question 20

```
public static long question20(ArrayList<Integer> list) {
    String answer = "";
    System.out.printf("Q20: O(%s)%n", answer);

    long count = 0;

    for (Integer value : list) {
        if (list.contains(value)) {
            count++;
        }
    }

    return count;
}
```

Question 21

```
public static long question21(int[] a) {
```

```
    String answer = "";
    System.out.printf("Q21: O(%s)%n", answer);

    long sum = 0;

    for (int i = 0; i < a.length; i++) {
        for (int j = 0; j < i; j++) {
            for (int k = 0; k < j; k++) {
                sum++;
            }
        }
    }

    return sum;
}
```

Question 22

```
public static long question22(int[] a) {
    String answer = "";
    System.out.printf("Q22: O(%s)%n", answer);

    long sum = 0;
    int size = a.length;

    while (size > 0) {
        for (int value : a) {
            sum += value;
        }

        size /= 2;
    }

    return sum;
}
```

Question 23

```
public static long question23(int[] a, int[] b) {
    String answer = "";
    System.out.printf("Q23: O(%s)%n", answer);

    long sum = 0;

    for (int x : a) {
        sum += x;
    }

    for (int y : b) {
        sum += y;
    }

    for (int x : a) {
        sum += x;
    }
```

```
    return sum;
}
```

Question 24

```
public static long question24(int[] a) {
    String answer = "";
    System.out.printf("Q24: O(%s)%n", answer);

    long sum = 0;

    for (int i = 0; i < a.length; i++) {
        int index = a.length - 1;

        while (index > 0) {
            sum += a[index];
            index /= 2;
        }
    }

    return sum;
}
```

Question 25

```
public static long question25(int[] a) {
    String answer = "";
    System.out.printf("Q25: O(%s)%n", answer);

    return fibonacciLike(a.length);
}

private static long fibonacciLike(int n) {
    if (n <= 1) {
        return n;
    }

    long previous = fibonacciLike(n - 1);
    long beforePrevious = fibonacciLike(n - 2);

    return previous + beforePrevious;
}
```

Question 26

```
public static long question26(int[] a) {
    String answer = "";
    System.out.printf("Q26: O(%s)%n", answer);

    long sum = 0;

    for (int i = 0; i < a.length; i++) {
        for (int j = 0; j < a.length; j++) {
            for (int k = 0; k < 5; k++) {
                sum += a[i] + a[j];
            }
```

```
        }
    }

    return sum;
}
```

Question 27

```
public static long question27(int[] a) {
    String answer = "";
    System.out.printf("Q27: O(%s)%n", answer);

    long sum = 0;

    for (int i = 0; i < a.length; i++) {
        sum += a[i];
    }

    for (int i = 0; i < a.length; i++) {
        sum += a[i];
    }

    return sum;
}
```

Question 28

```
public static long question28(int[] a, int[] b) {
    String answer = "";
    System.out.printf("Q28: O(%s)%n", answer);

    long count = 0;

    for (int x : a) {
        for (int y : b) {
            if (x == y) {
                count++;
            }
        }
    }

    return count;
}
```

Question 29

```
public static long question29(int[] a) {
    String answer = "";
    System.out.printf("Q29: O(%s)%n", answer);

    long sum = 0;

    for (int i = 0; i < a.length; i++) {
        if (a[i] % 2 == 0) {
            sum += a[i];
        } else {
```

```
                sum -= a[i];
            }
        }

        return sum;
    }
```

Question 30

```
    public static long question30(int[] a) {
        String answer = "";
        System.out.printf("Q30: O(%s)%n", answer);

        if (a.length < 3) {
            return a.length;
        }

        int middle = a.length / 2;

        return a[0] + a[middle] + a[a.length - 1];
    }
}
```

End of worksheet. No answer key is included in this PDF.